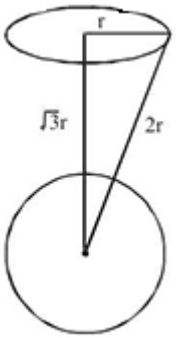
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A uniform ring of mass m and radius r is placed directly above a uniform sphere of mass M and of equal radius. The centre of the ring is directly above the centre of the sphere at a distance $r\sqrt{3}$ as shown in the figure. The gravitational force exerted by the sphere on the ring will be



(A) $\frac{\mathrm{GMm}}{8\mathrm{r}^2}$

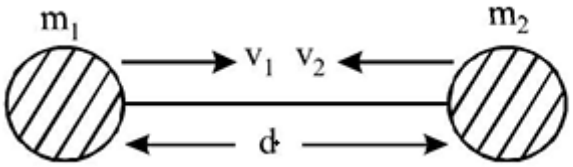
(B) $\frac{\mathrm{GMm}}{4\mathrm{r}^2}$

(C) $\sqrt{3}\frac{\mathrm{GMm}}{8\mathrm{r}^2}$

(D) $\frac{\mathrm{GMm}}{8\mathrm{r}^3\sqrt{3}}$

Q.2

Two hypothetical planets of masses m_1 and m_2 are at rest when they are infinite distance apart. Because of the gravitational force they move towards each other along the line joining their centres.
What is their speed when their separation is 'd'? (Speed of m_1 is v_1 and that of m_2 is v_2)



(A) $\mathrm{v}_1 = \mathrm{v}_2$

(B) $v_1 = m_2\sqrt{\frac{2G}{d(m_1+m_2)}}, v_2 = m_1\sqrt{\frac{2G}{d(m_1+m_2)}}$

(C) $\mathrm{v}_1 = \mathrm{m}_1\sqrt{\frac{2\mathrm{G}}{\mathrm{d}(\mathrm{m}_1+\mathrm{m}_2)}}, \mathrm{v}_2 = \mathrm{m}_2\sqrt{\frac{2\mathrm{G}}{\mathrm{d}(\mathrm{m}_1+\mathrm{m}_2)}}$

(D) $\mathrm{v}_1 = \mathrm{m}_2\sqrt{\frac{2\mathrm{G}}{\mathrm{m}_1}}, \mathrm{v}_2 = \mathrm{m}_2\sqrt{\frac{2\mathrm{G}}{\mathrm{m}_2}}$

Q.3

If the radius of the earth were to shrink by 1%, with its mass remaining the same, the acceleration due to gravity on the earth's surface would increase by $x\%$. Then find the value of x.

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(A) 1

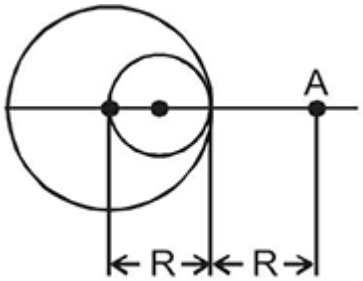
(B) 2

(C) 3

(D) 4

Q.4

A solid sphere of uniform density and radius R applies a gravitational force of attraction equal to F_1 on a particle placed at A, distance $2R$ from the centre of the sphere.



A spherical cavity of radius $\mathrm{R}/2$ is now made in the sphere as shown in the figure. The sphere with cavity now applies a gravitational force F_2 on the same particle placed at A. Then what is the value of ratio $\mathrm{F}_2/\mathrm{F}_1$?

(A) 0.122

(B) 0.115

(C) 0.777

(D) 0.155

Q.5

If suddenly the gravitational force of attraction between Earth and a body revolving around it becomes zero, then the body will

(A) continue to move in its orbit with same velocity

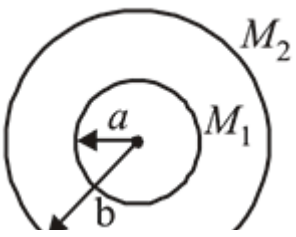
(B) move tangentially to the original orbit with same velocity

(C) become stationary in its orbit

(D) move towards the earth

Q.6

Two concentric uniform shells of mass M_1 and M_2 are as shown in the figure. A particle of mass m is located just within the shell M_2 on its inner surface. Gravitational force on ' m ' due to M_1 and M_2 will be



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(A) zero

(B) $\frac{GM_1m}{b^2}$

(C) $\frac{G(M_1+M_2)m}{b^2}$

(D) None of these

Q.7

For a particle inside a uniform spherical shell, the gravitational force on the particle is

(A) infinite

(B) zero

(C) $\frac{-Gm_1m_2}{r^2}$

(D) $\frac{Gm_1\ m_2}{r^2}$

Q.8

Two spheres of masses m and M are situated in air and the gravitational force between them is F. The space around the masses is now filled with a liquid of specific gravity 3 . The gravitational force will now be

(A) $\frac{F}{9}$

(B) $3F$

(C) F

(D) $\frac{F}{3}$

Q.9

A mass M is split into two parts m and $(M-m)$, which are then separated by a certain distance. What ratio of m/M maximizes the gravitational force between the two parts?

(A) $\frac{1}{3}$

(B) $\frac{1}{2}$

(C) $\frac{1}{4}$

(D) $\frac{1}{5}$

Q.10

Two stars of mass m_1 and m_2 are parts of a binary system. The radii of their orbits are r_1 and r_2 respectively, measured from the C.M. of the system. The magnitude of gravitational force m_1 exerts on m_2 is

(A) $\frac{m_1m_2G}{(r_1+r_2)^2}$

(B) $\frac{m_2G}{(r_1+r_2)^2}$

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(C) $\frac{m_1 G}{(r_1+r_2)^2}$

(D) $\frac{(m_1+m_2)}{(r_1+r_2)^2}$

Q.11

Two bodies of masses $4\ \text{kg}$ and $9\ \text{kg}$ are separated by a distance of $60\ \text{cm}$. A $1\ \text{kg}$ mass is placed in between these two masses. If the net force on 1kg is zero, then its distance from $4\ \text{kg}$ mass is

(A) $26\ \text{cm}$

(B) $30\ \text{cm}$

(C) $28\ \text{cm}$

(D) $24\ \text{cm}$

Q.12

If masses of two point objects is doubled and distance between them is tripled, then gravitational force of attraction between them will nearly

(A) increase by 225%

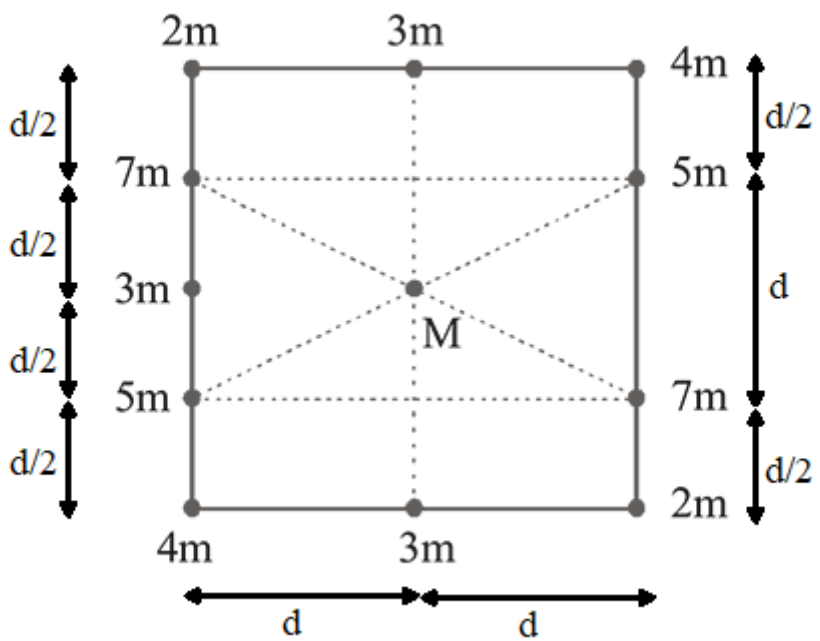
(B) decrease by 44%

(C) decrease by 56%

(D) increase by 125%

Q.13

A central particle M is surrounded by a square array of other particles, separated by either distance d or distance $d/2$ along the perimeter of the square. The magnitude of the gravitational force on the central particle due to the other particles is



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(A) $\frac{9GMm}{d^2}$

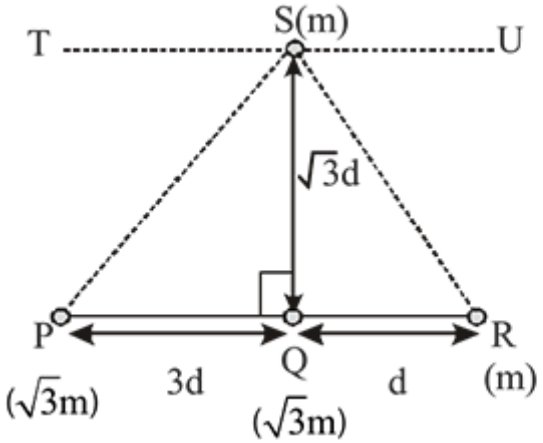
(B) $\frac{5GMm}{d^2}$

(C) $\frac{3GMm}{d^2}$

(D) $\frac{GMm}{d^2}$

Q.14

Three particles P, Q and R are placed as per given figure. Masses of P, Q and R are $\sqrt{3}$ m, $\sqrt{3}$ m and m respectively. The gravitational force on a fourth particle S of mass m is equal to



(A) $\frac{\sqrt{3}GM^2}{2\,d^2}$ in ST direction only

(B) $\frac{\sqrt{3}GM^2}{2\,d^2}$ in SQ direction and $\frac{\sqrt{3}GM^2}{2\,d^2}$ in SU direction

(C) $\frac{\sqrt{3}GM^2}{2\,d^2}$ in SQ direction only

(D) $\frac{\sqrt{3}GM^2}{2\,d^2}$ in SQ direction and $\frac{\sqrt{3}GM^2}{2\,d^2}$ in ST direction

Q.15

A man goes to a height equal to the radius of the earth from its surface. The weight of the person at that height relative to his weight on the surface of earth is

(A) $\frac{1}{2}$

(B) $\frac{1}{3}$

(C) $\frac{1}{4}$

(D) $\frac{1}{5}$

Answers & Solutions

Q.1 Answer:

$\sqrt{3}\frac{GMm}{8r^2}$

Solution:

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On a differential part of ring,

the gravitational force (dF) **by sphere** $= \frac{\mathrm{GMdm}(2)\overrightarrow{\mathrm{r}}}{(2\mathrm{R}^2)(2\mathrm{R})}$

$\Rightarrow \mathrm{d}\overrightarrow{\mathrm{F}} = \frac{4\mathrm{Mdm}\,\overrightarrow{\mathrm{r}}}{4\mathrm{R}^2(\mathrm{R})} = \frac{\mathrm{GMdm}\,\overrightarrow{\mathrm{r}}}{4\mathrm{R}^3}$

Total gravitational $(\mathrm{F}) = \int_0^{\mathrm{m}} \frac{\mathrm{GMdm}\,\overrightarrow{\mathrm{r}}}{4\mathrm{R}^3}$

$\overrightarrow{\mathrm{r}}$ **is resolved into vectors are with magnitude** $\left(\frac{\sqrt{3}\vec{r}}{2}\right)$

directed perpendicular to the plane of ring and other radially with magnitude $\left(\frac{|\vec{r}|}{2}\right)$

Let them be $\mathrm{r}_\perp$ **and** $\overrightarrow{\mathrm{r}}_\mathrm{c}$ **receptively.**

Then,

$\overrightarrow{\mathrm{r}} = \overrightarrow{\mathrm{r}_\perp} + \overrightarrow{\mathrm{rc}}$

$$\text{So, dF} = \frac{\mathrm{GMdm}\overrightarrow{r^2}}{4\mathrm{R}^3} = \frac{\mathrm{GMdm}}{4\mathrm{R}^3}\left(\overrightarrow{\mathrm{r}_\perp} + \overrightarrow{\mathrm{r}_\mathrm{c}}\right)$$

$$\overrightarrow{\mathrm{F}} = \int_0^{\mathrm{m}} \overrightarrow{\mathrm{df}} = \int_0^{\mathrm{m}} \frac{\mathrm{GMdm}\left(\overrightarrow{\mathrm{r}_\perp}\right)}{4\mathrm{R}^3} + \int_0^{\mathrm{m}} \frac{\mathrm{GMdm}\,\overrightarrow{\mathrm{r}}_\mathrm{c}}{4\mathrm{R}^3}$$

$$\overrightarrow{\mathrm{F}} = \int_0^{\mathrm{m}} \frac{\mathrm{GMdm}}{4\mathrm{R}^3}\left(\frac{\sqrt{3}}{2}\right)\mathrm{Rr}_\perp + \int_0^{\mathrm{m}} \frac{\mathrm{GMdm}}{4\mathrm{R}^3}\left(\frac{\mathrm{R}}{2}\right)\mathrm{r}_\mathrm{c}$$

$\overrightarrow{\mathrm{F}} = \frac{\sqrt{3}\mathrm{GM}}{8\mathrm{R}^2}\int_0^{\mathrm{m}} \mathrm{dmr}_\perp + \frac{\mathrm{GM}}{8\mathrm{R}^2}\int_0^{\mathrm{m}} \mathrm{dmr}_\mathrm{c}$

At every point $r_\perp$ **is same but**

r_c **changes accordingly such that**

$\int_0^{\mathrm{m}} \mathrm{dmr}_\perp = \mathrm{Mr} \perp$ **and** $\int_0^{\mathrm{m}} \mathrm{r}_\mathrm{c} = 0$

So, $\vec{F} = \frac{\sqrt{3}\mathrm{GM}}{8\mathrm{R}^2}\mathrm{mr}_\perp$

option C **is correct.**

Q.2 Answer:

$v_1 = m_2\sqrt{\frac{2G}{d(m_1+m_2)}}, v_2 = m_1\sqrt{\frac{2G}{d(m_1+m_2)}}$

Solution:

We choose reference point, infinity, where total energy of the system is zero. So, initial energy of the system $= 0$ **Final energy** $= \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 - \frac{Gm_1m_2}{d}$

From conservation of energy, Initial energy = Final energy $\therefore 0 = \frac{1}{2}\,\mathrm{m}_1\mathrm{v}_1^2 + \frac{1}{2}\,\mathrm{m}_2{\mathrm{v}_2}^2 - \frac{\mathrm{Gm}_1\,\mathrm{m}_2}{\mathrm{d}}$

or $\frac{1}{2}\,\mathrm{m}_1\mathrm{v}_1^2 + \frac{1}{2}\,\mathrm{m}_1\mathrm{v}_2^2 = \frac{\mathrm{Gm}_1\,\mathrm{m}_2}{\mathrm{d}}$

By conservation of linear momentum

$\mathrm{m}_1\mathrm{v}_1 + \mathrm{m}_2\mathrm{v}_2 = 0$ **or** $\frac{\mathrm{v}_1}{\mathrm{v}_2} = -\frac{\mathrm{m}_2}{\mathrm{m}_1} \Rightarrow \mathrm{v}_2 = -\frac{\mathrm{m}_1}{\mathrm{m}_2}\mathrm{v}_1$

Putting value of v_2 **in equation (1), we get** $\mathrm{m}_1\mathrm{v}_1^2 + \mathrm{m}_2\left(-\frac{\mathrm{m}_1\mathrm{v}_1}{\mathrm{m}_2}\right)^2 = \frac{2\mathrm{Gm}_1\,\mathrm{m}_2}{\mathrm{d}}$

$\frac{\mathrm{m}_1\,\mathrm{m}_2\mathrm{v}_1^2 + \mathrm{m}_1^2\mathrm{v}_1^2}{\mathrm{m}_2} = \frac{2\mathrm{Gm}_1\,\mathrm{m}_2}{\mathrm{d}}$

$\mathrm{v}_1 = \sqrt{\frac{2\mathrm{Gm}_2^2}{\mathrm{d}(\,\mathrm{m}_1+\mathrm{m}_2)}} = \mathrm{m}_2\sqrt{\frac{2\mathrm{G}}{\mathrm{d}(\mathrm{m}_1+\mathrm{m}_2)}}$

Similarly $\mathrm{v}_2 = -\mathrm{m}_1\sqrt{\frac{2\mathrm{G}}{\mathrm{d}(\mathrm{m}_1+\mathrm{m}_2)}}$

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Q.3 Answer:

2

Solution:

Acceleration due to gravity on earth's surface $\mathrm{g}=\mathrm{G}\frac{\mathrm{M}}{\mathrm{R}^2}$
This implies that as radius decreases, the acceleration due to gravity increases. $\frac{\Delta \mathrm{g}}{\mathrm{g}}=-2\frac{\Delta \mathrm{R}}{\mathrm{R}}$ **But** $\frac{\Delta \mathrm{R}}{\mathrm{R}}=-1\%$
('- sign is due to shrinking of earth) $\therefore \quad \frac{\Delta \mathrm{g}}{\mathrm{g}}=-2\times(-1\%)=2\%$

Q.4 Answer:

0.777

Solution:

$C \rightarrow$ **cavity,** $T \rightarrow$ **Total,** $R \rightarrow$ **remaining**
$F_1=\frac{GMm}{(2R)^2}$ **... (i)**
$F_R=F_2=F_T-F_C=F_1-F_C$
or $F_2=\frac{GMm}{(2R)^2}-\frac{G\left(\frac{M}{8}\right)m}{(3R/2)^2}$
or $F_2=\frac{14GMm}{72R^2}$
From Eqs. (i) and (ii) we get,
$\frac{F_2}{F_1}=\frac{7}{9}=0.777$

Q.5 Answer:

move tangentially to the original orbit with same velocity

Solution:

Bodywill move tangentially to the original orbit with same velocity.

Q.6 Answer:

$\frac{\mathrm{G}M_1m}{\mathrm{b}^2}$

Solution:

Gravitational force will be due to M_1 **only.**

Q.7 Answer:

zero

Solution:

Various regions of spherical shell attract the point mass inside it in various directions. These forces cancel each other completely. Therefore the gravitational force on the particle is zero.

Q.8 Answer:

F

Solution:

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Gravitational force is independent of medium. Hence, this will remain same.

Q.9 Answer:

$\frac{1}{2}$

Solution:

$F = G\frac{m(M-m)}{r^2}$

For maximum value of F, $dF/dm = 0$, **and so** $m = M/2$

Q.10 Answer:

$\frac{m_1 m_2 G}{(r_1+r_2)^2}$

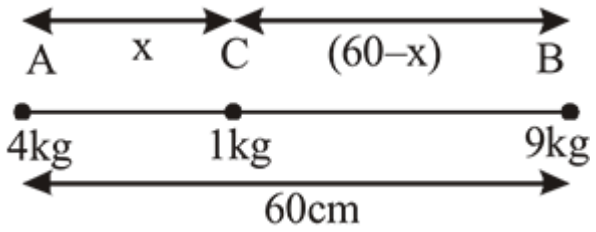
Solution:

$F = \frac{Gm_1m_2}{r^2} = \frac{Gm_1m_2}{(r_1+r_2)^2}$

Q.11 Answer:

24 cm

Solution:



$\therefore \quad G\frac{4\times1}{x^2} = G\frac{9\times1}{(60-x)^2}$

or $\frac{2}{3} = \frac{x}{(60-x)} \Rightarrow x = 24$ cm

Q.12 Answer:

decrease by 56%

Solution:

$\%$ **change** $= \frac{\left[\frac{G(2m_1)(2m_2)}{(3r)^2} - \frac{Gm_1m_2}{r^2}\right]}{\frac{Gm_1m_2}{r^2}} \times 100$

$= \frac{\frac{4}{9}-1}{1} \times 100 = -56\%$

- ve sign indicates that force of attraction decreases.

Q.13 Answer:

$\frac{3GMm}{d^2}$

Solution:

$F = \frac{GM(3m)}{d^2} = \frac{3GMm}{d^2}$

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Q.14 Answer:

$\frac{\sqrt{3}GM^2}{2\,d^2}$ in SQ direction only

Solution:

In horizontal direction Net force $= \frac{G\sqrt{3}\,mm}{12\,d^2}\cos 30^\circ - \frac{Gm^2}{4\,d^2}\cos 60^\circ$

$= \frac{Gm^2}{8\,d^2} - \frac{Gm^2}{8\,d^2} = 0$

In vertical direction, Net force $= \frac{G\sqrt{3}m^2}{12d^2}\cos 60^\circ + \frac{G\sqrt{3}m^2}{3d^2} + \frac{Gm^2}{4d}\cos 30^\circ$

$= \frac{\sqrt{3}Gm^2}{24d^2} + \frac{\sqrt{3}Gm^2}{3d^2} + \frac{\sqrt{3}Gm^2}{8d^2}$

$= \frac{\sqrt{3}Gm^2}{d^2}\left[\frac{1+8+3}{24}\right] = \frac{\sqrt{3}Gm^2}{2d^2}$ **along SQ.**

Q.15 Answer:

$\frac{1}{4}$

Solution:

At the surface of the earth, the weight of the person.

$W = mg \Rightarrow \frac{mGM}{R^2}$

At a height h

weight $W' = mg' \Rightarrow \frac{mGM}{(R+h)^2}$

$\therefore \frac{W'}{W} = \frac{R^2}{(R+h)^2}$

When $h = R$, $\frac{W'}{W} = \frac{R^2}{(2R)^2} = \frac{1}{4}$

So, weight becomes one-forth.

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